

PMU Event Filtering for Confidential Guests

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Agenda

PMU Event Filtering overview

SW based filtering

HW Assisted Guest Event filtering

Challenges with Confidential Guests

Proposed solution



PMU Event Filtering: Overview

A guest may profile itself, but only on the events the host is willing to let it measure.

Guest (VM)

runs perf, asks for

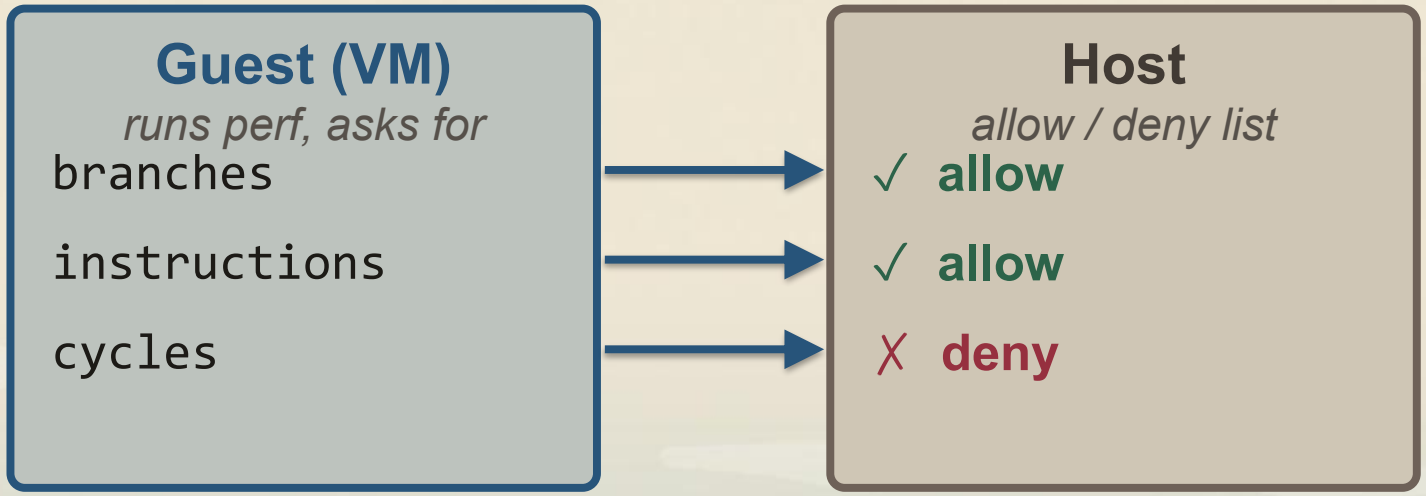
```
branches

instructions

cycles
```

PMU Event Filtering: Overview

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Allow the safe events, deny the rest - the host does not trust the guest.

Software based Event filtering

Apply the policy

VMM

owns the policy

branches	allow
instructions	allow
cycles	deny

KVM_SET_PMU_EVENT_FILTER

KVM registers the VMM Policy

branches	allow
instructions	allow
cycles	deny

Use the policy

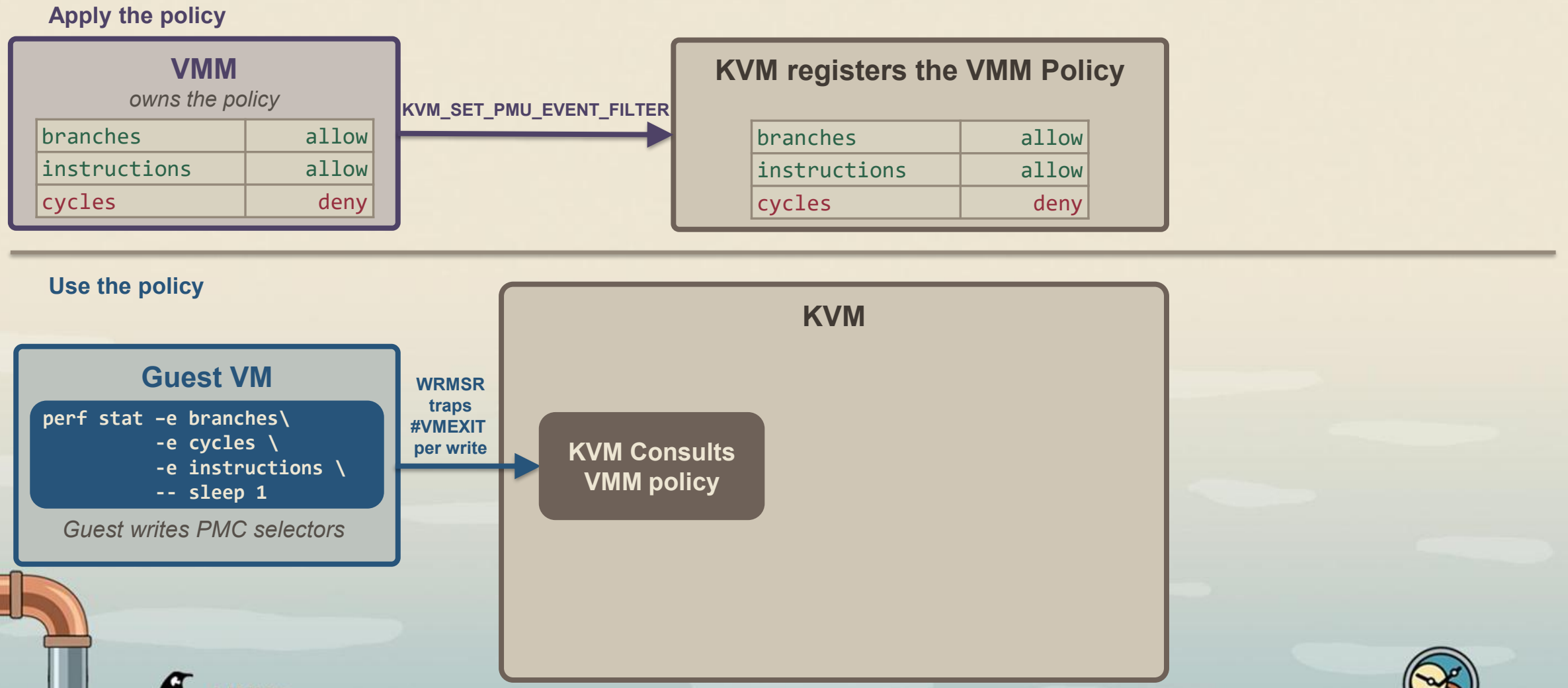
Guest VM

```
perf stat -e branches\
          -e cycles \
          -e instructions \
          -- sleep 1
```

Guest writes PMC selectors

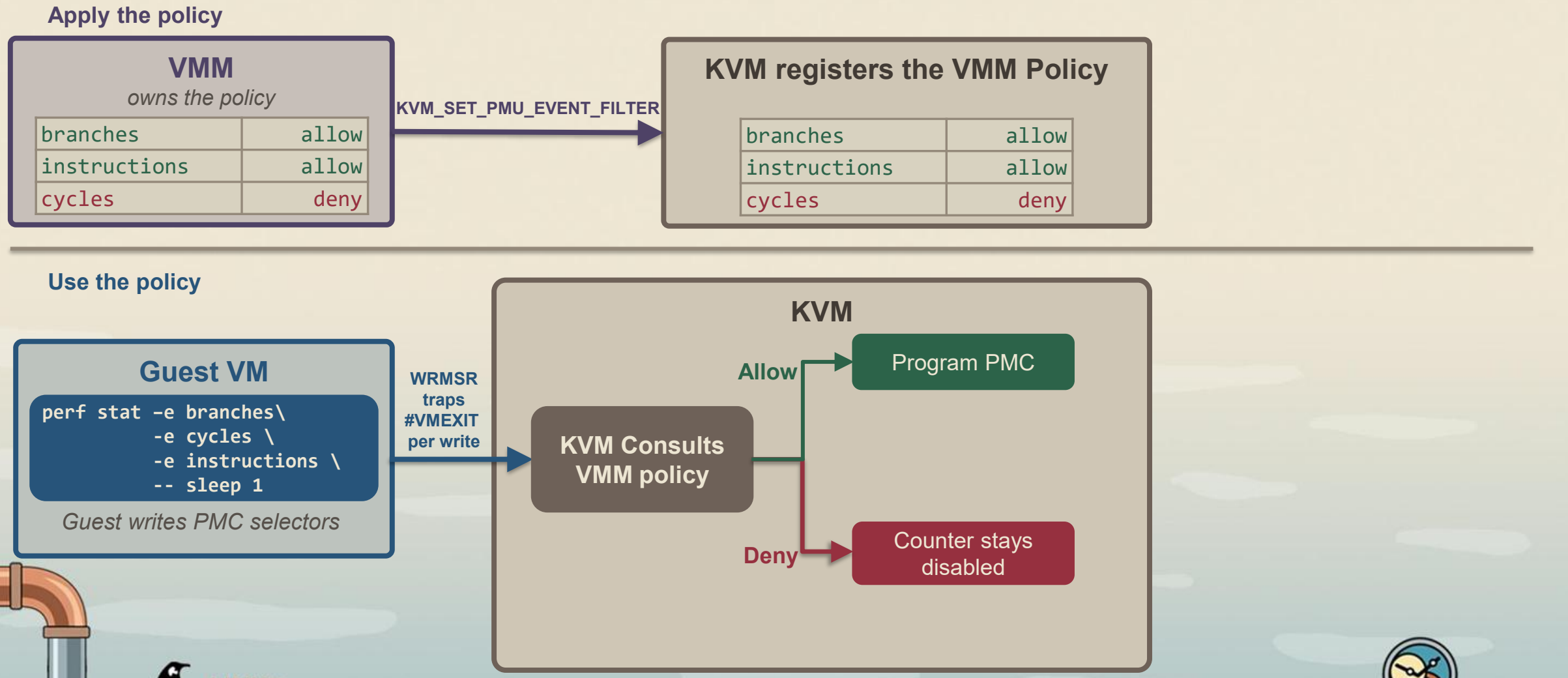


Software based Event filtering



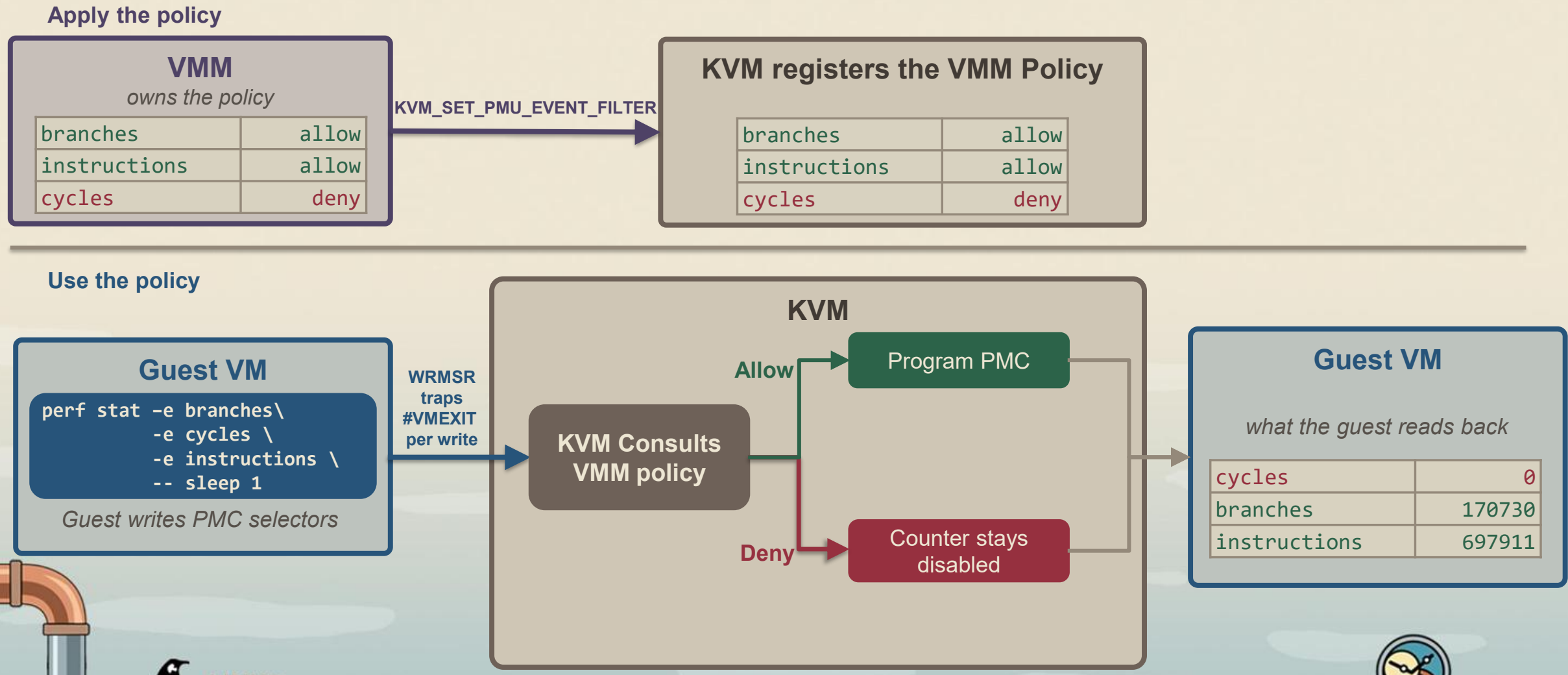
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Software based Event filtering



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Software based Event filtering



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Enabling Guest Event Filtering

What the host owns

PMC virtualization
+
PMC Event Filtering

VMCB Control Area

Allowed list
`ALLOWED_PERF_CTL[0..5]`
`ALLOWED_PERF_CTL_MASK`

What hardware does with it

Enabling an event (X) - WRMSR PerfEvtSeln / WRMSR PerfCntGlobalCtl

guest writes `X` to `PerfEvtSeln`

`X & ALLOWED_PERF_CTL_MASK == ALLOWED_PERF_CTL[0/1/2/3/4/5] & ALLOWED_PERF_CTL_MASK`

✓ **ALLOW**
write completes - no exit

✗ **DENY**
the write is trapped - but where?

SVM	`VMEXIT_WRMSR`
SEV-ES / SEV-SNP	`#VC in the guest`

On every VMRUN, all enabled events should be listed in `ALLOW_PERF_CTL[0..5]`

Events in Allowed list: No Exit Needed

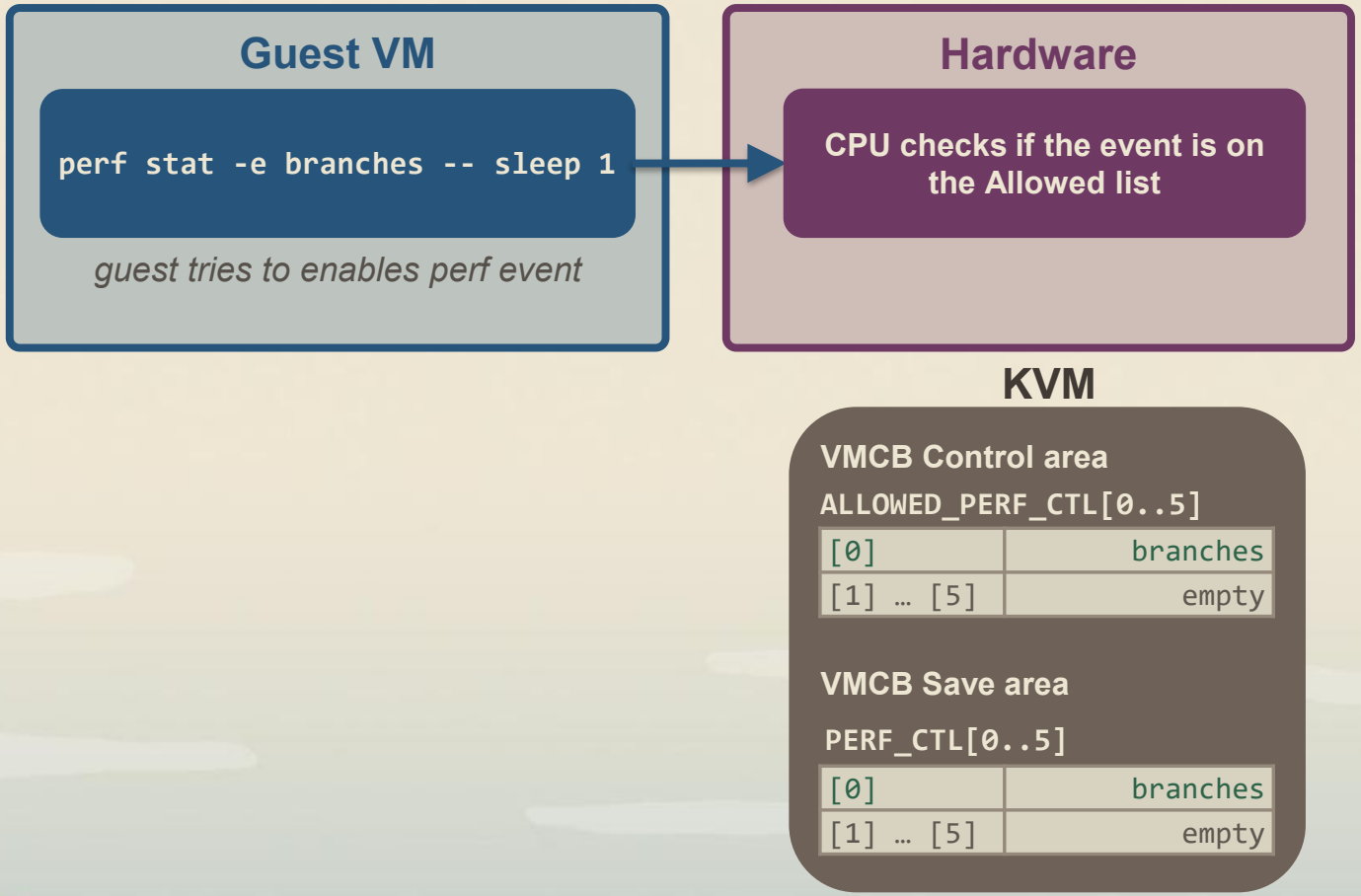
Guest VM

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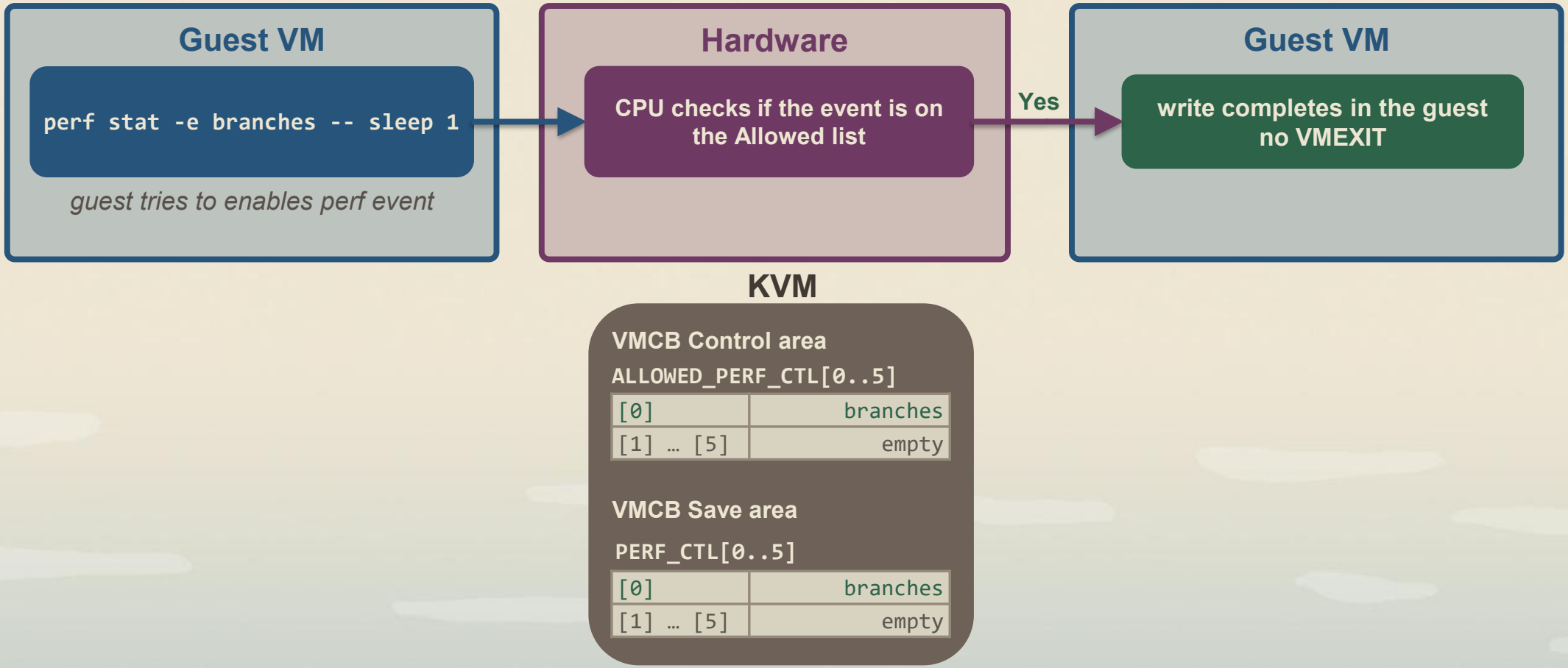
guest tries to enables perf event

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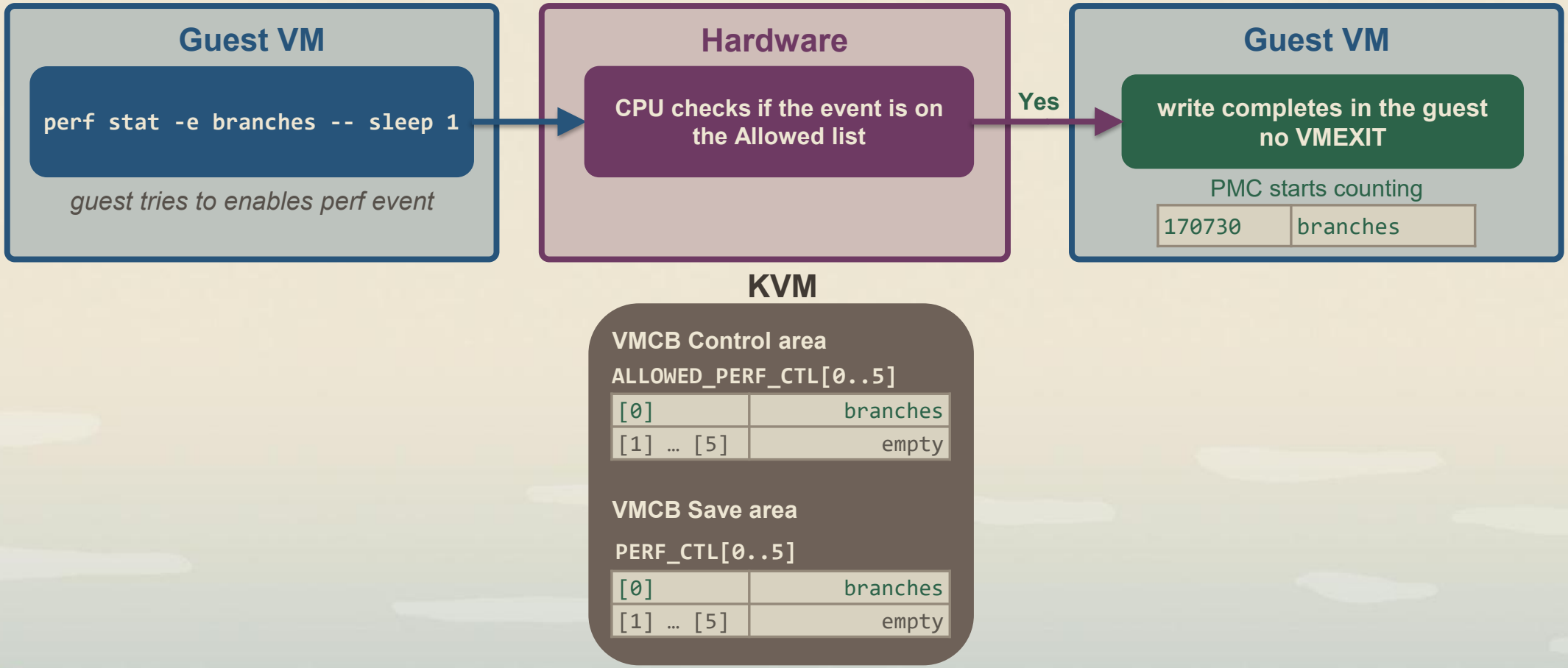
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Events in Allowed list: No Exit Needed



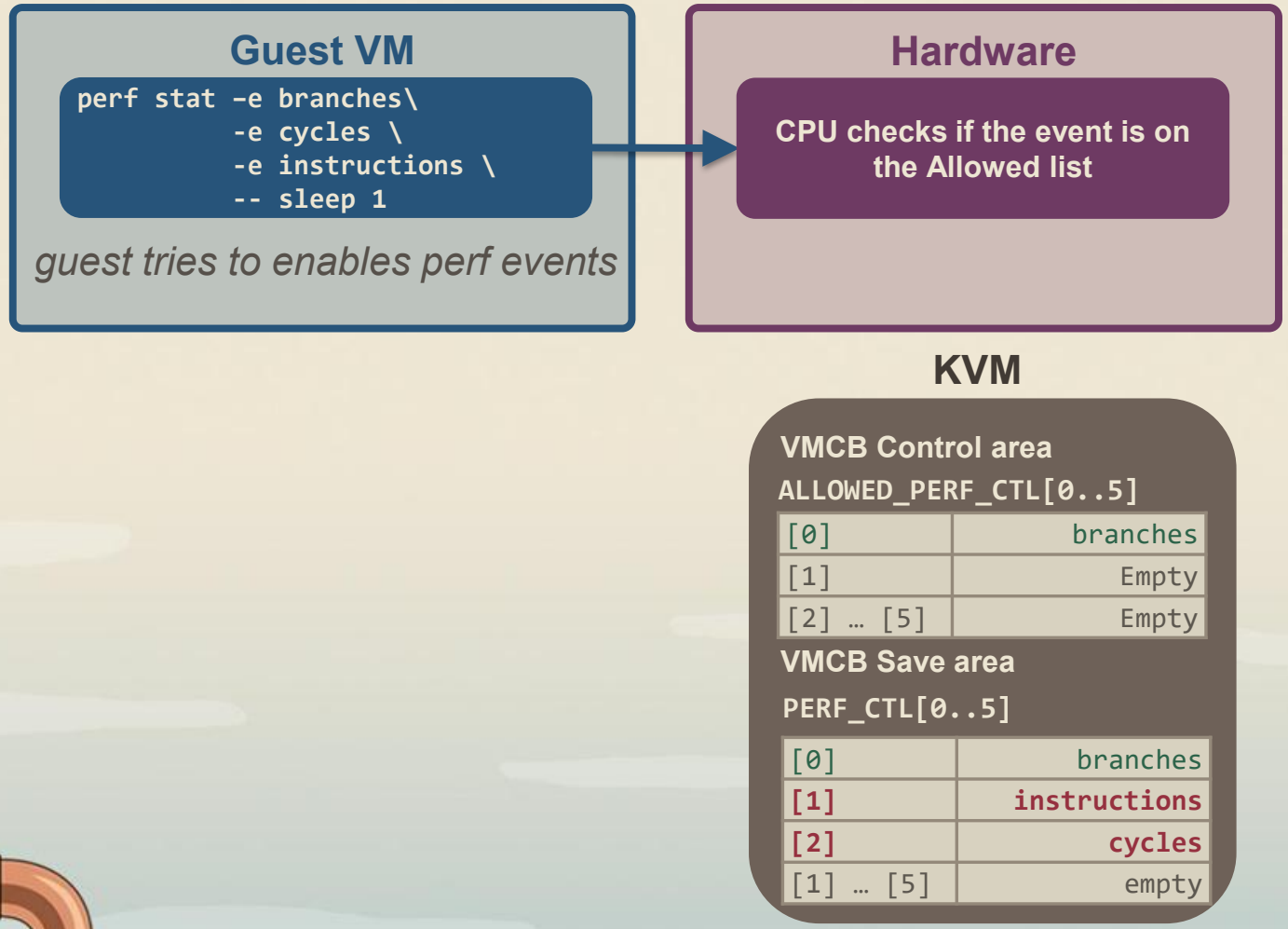
Events not in Allowed list: Exit to KVM

Guest VM

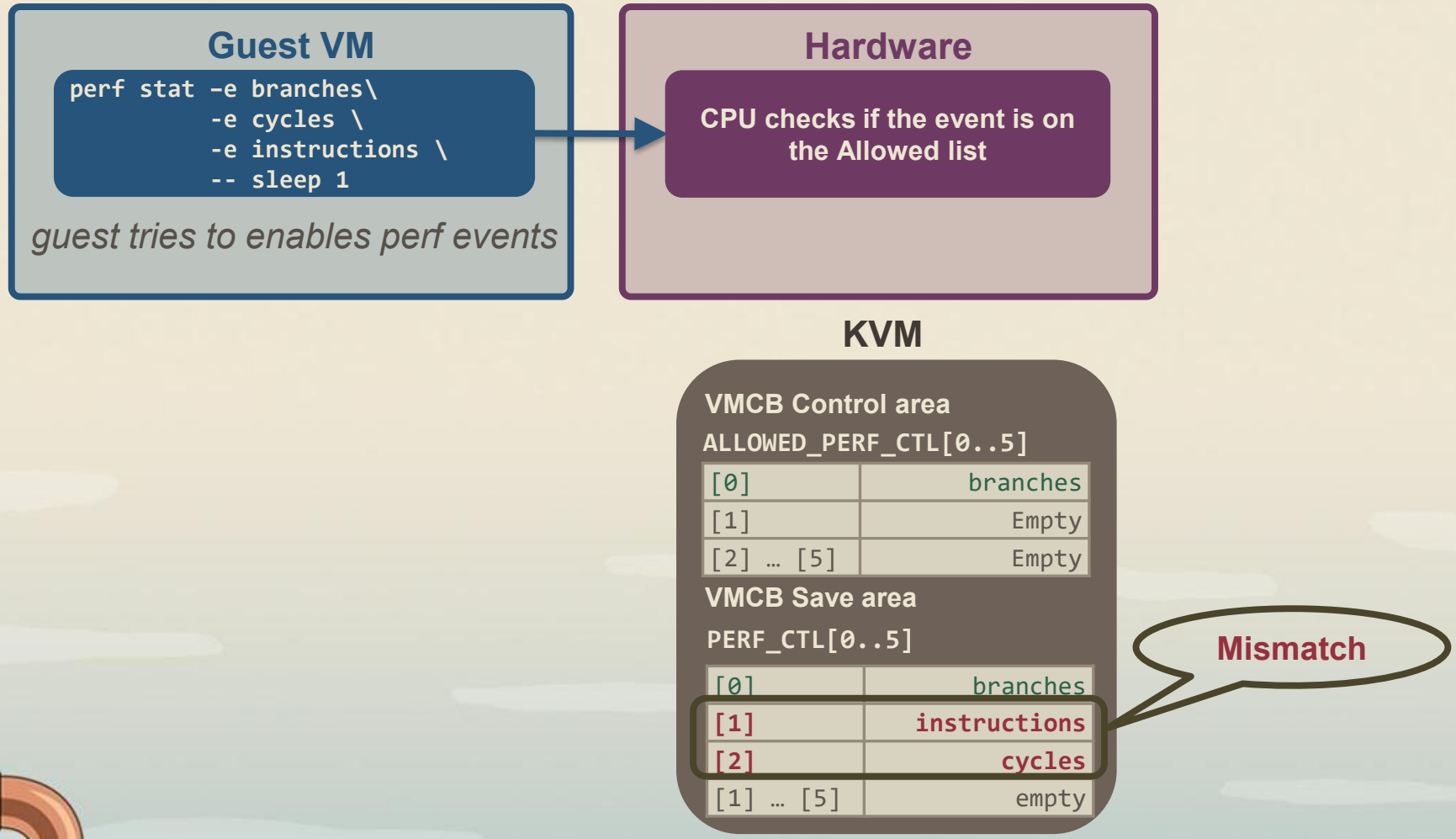
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          -- sleep 1
```

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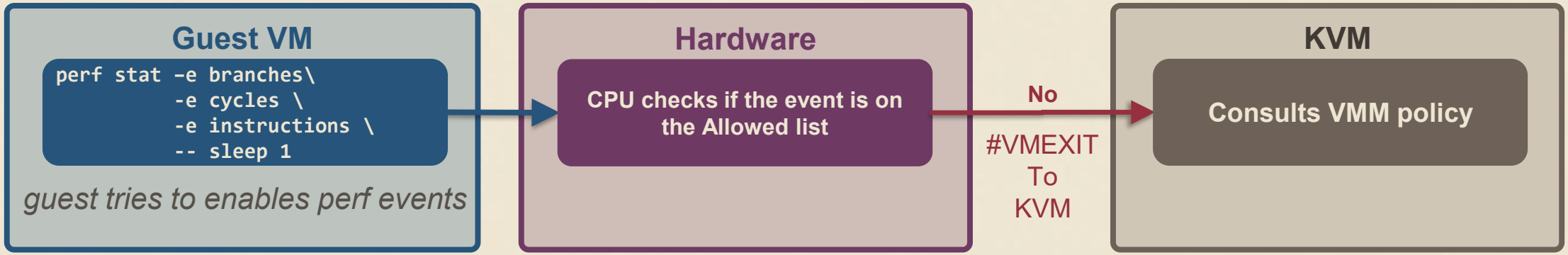
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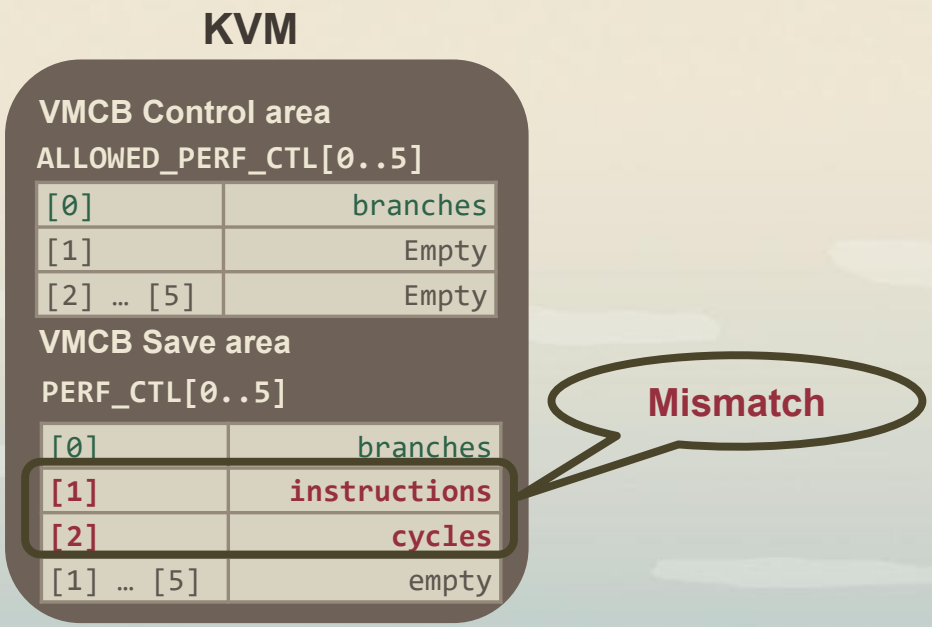


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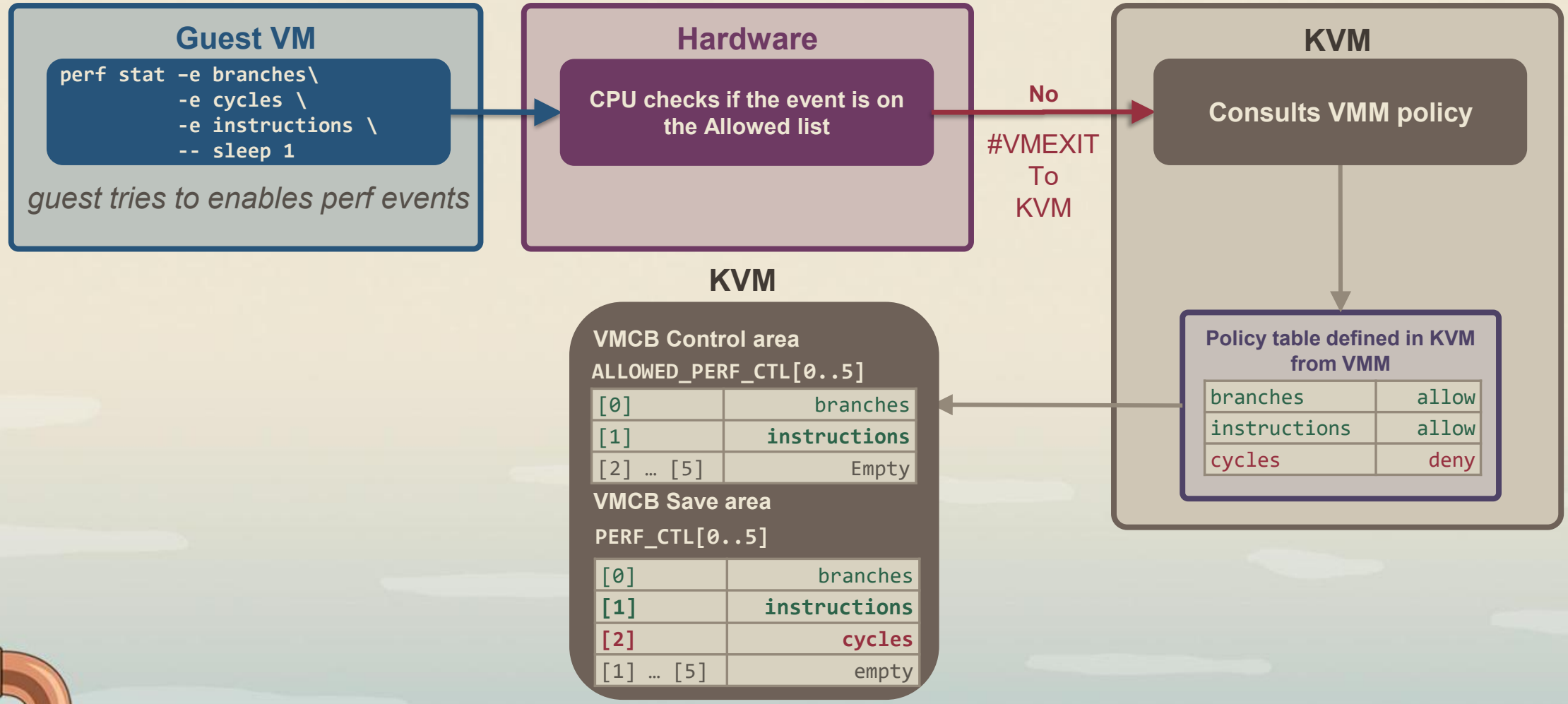


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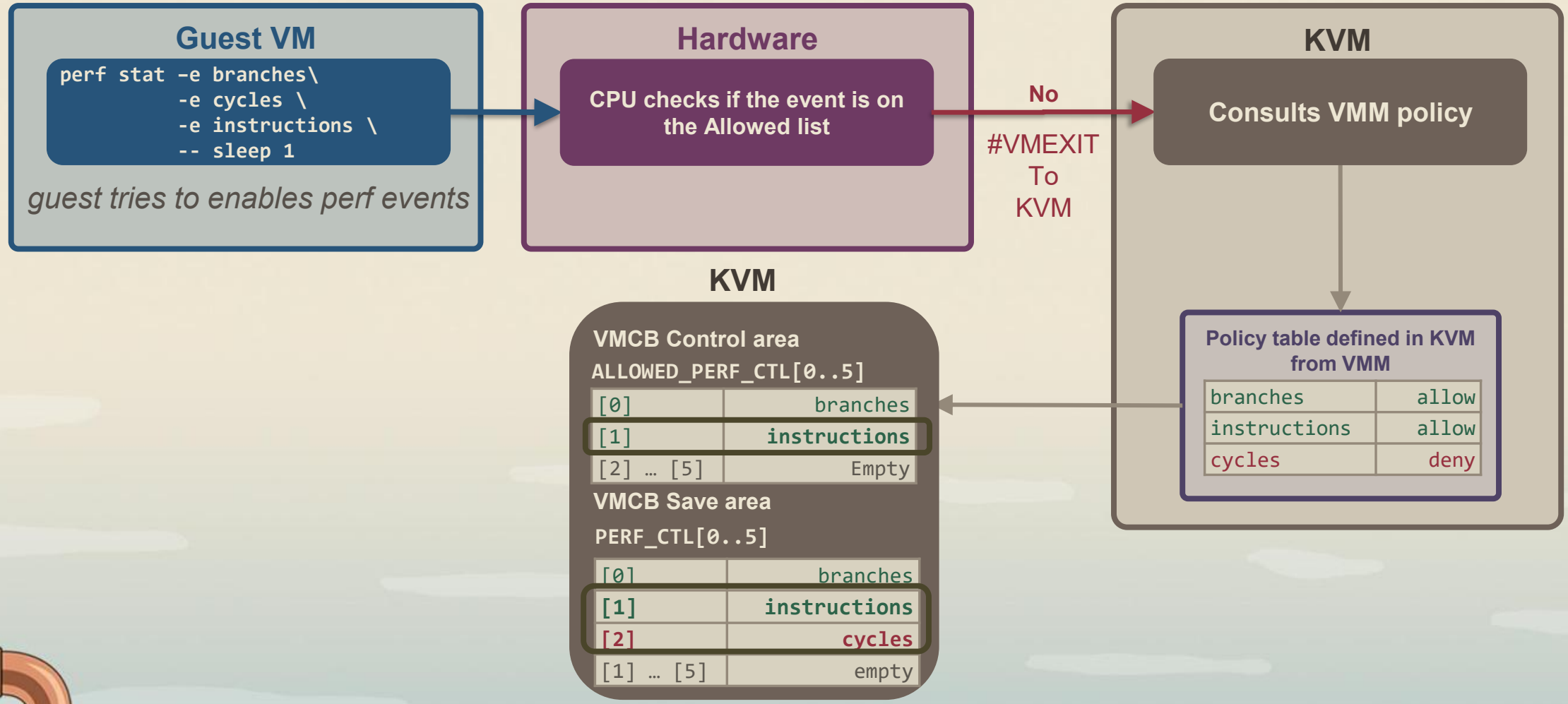




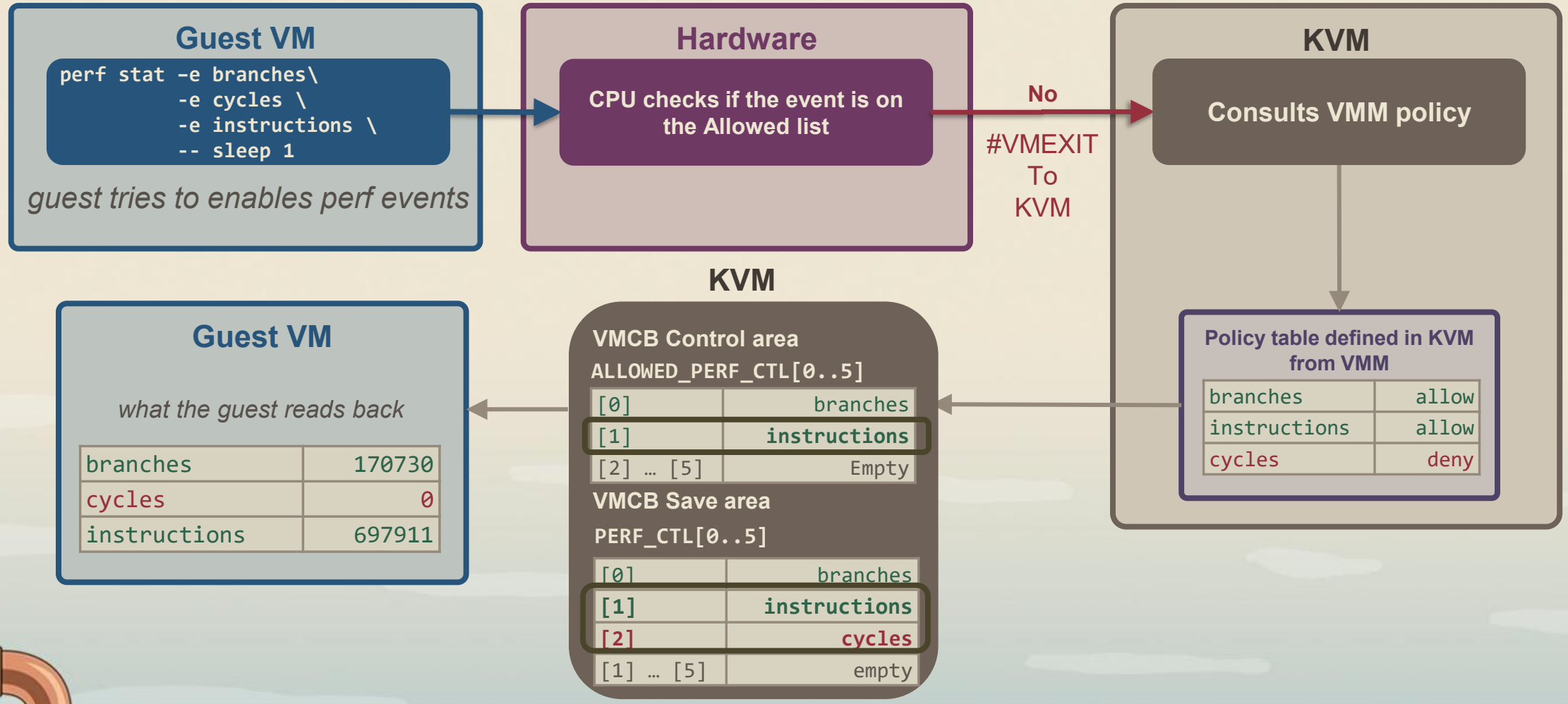
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Mediated PMU vs. VPMC vs. HW Assisted Guest Event filtering

```
Guest instance (single vCPU) – Perform a top-down analysis
$ sudo perf stat –M PipelineL1  -- sysbench --cpu-max-prime=1000 --threads=1 cpu run
```

```
Host instance – Count VM-Exits for hardware counter accesses
$ sudo perf stat –e kvm:kvm_msr --filter="((ecx >= 0xc0010000 && ecx <= 0xc0010007) || (ecx >=
0xc0010200 && ecx <= 0xc001020b) || (ecx >= 0xc0000300 && ecx <= 0xc0000303))"
```

101,528 kvm:kvm_msr	101,264 kvm:kvm_msr	7,503 kvm:kvm_msr
Mediated PMU No of Exits: 101,528	VPMC No of Exits: 101,264	HW-Assisted Guest Event filtering No of Exits: 7503

"VPMC and Mediated PMU will always intercept PERF_CTL MSRs irrespective of the event code and whether we are enabling or disabling the counters"



HW Assisted Guest Event filtering for Confidential Guests



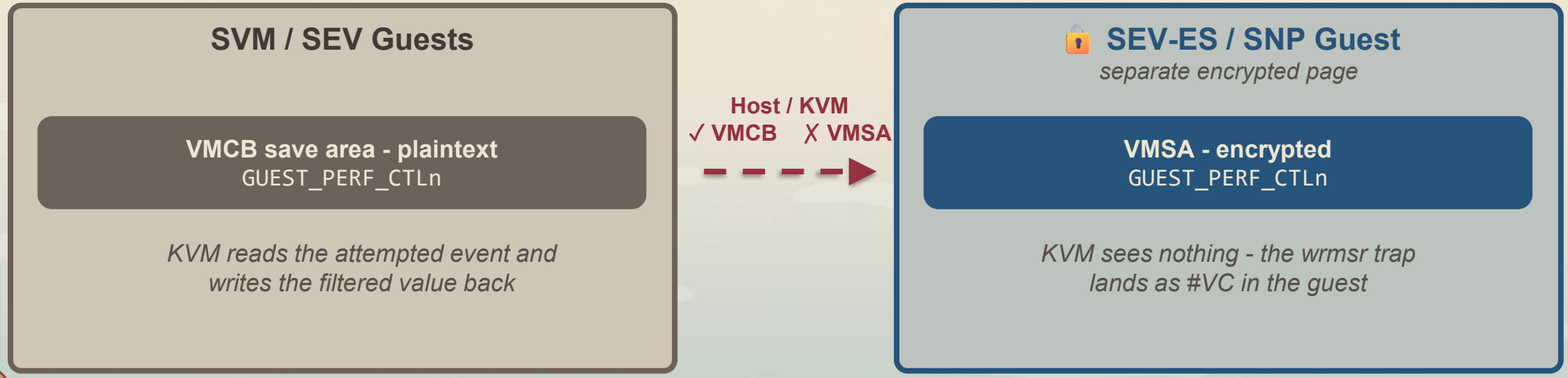
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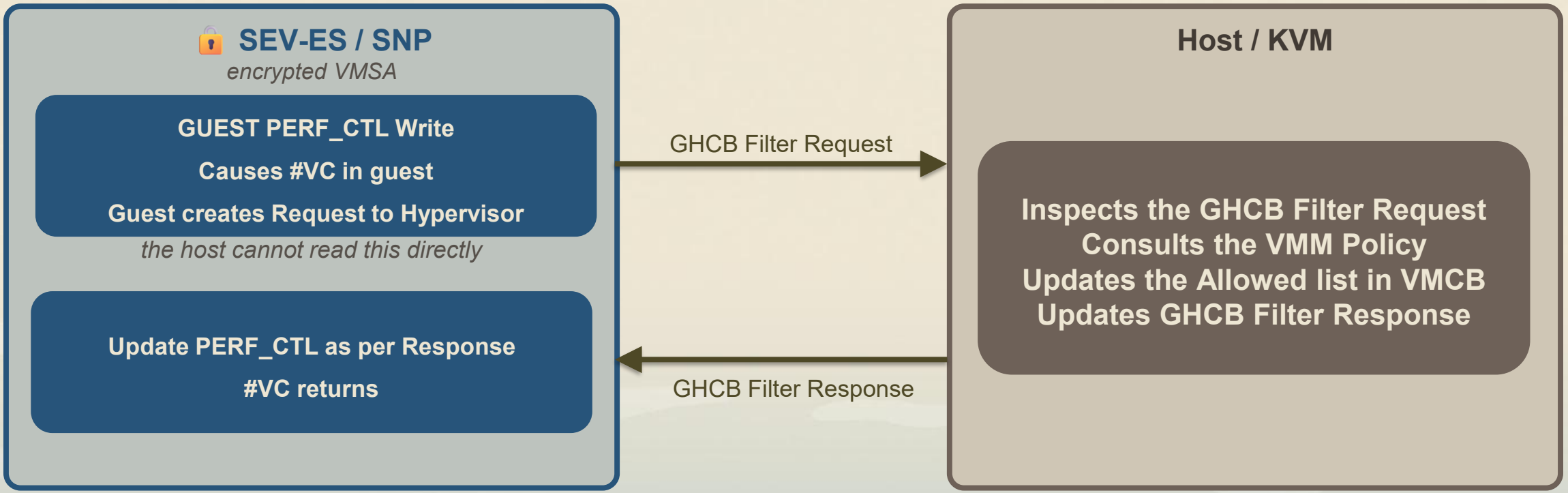
The Problem

SVM guest: VMCB save area is accessible to the hypervisor
SEV-ES / SNP: save area is the encrypted VMSA

Result: HW Assisted Guest Event filtering can't be used as is



The Idea


SEV-ES / SNP
encrypted VMSA
GUEST PERF_CTL Write
Causes #VC in guest
Guest creates Request to Hypervisor
the host cannot read this directly
Update PERF_CTL as per Response
#VC returns
GHCB Filter Request
GHCB Filter Response
Host / KVM
Inspects the GHCB Filter Request
Consults the VMM Policy
Updates the Allowed list in VMCB
Updates GHCB Filter Response

Events not in Allowed list: Exit to KVM

Guest VM

```
perf stat -e branches\
          -e cycles \
          -e instructions \
          -- sleep 1
```

Hardware

CPU checks if the event is on the Allowed list

🔒 VMSA (encrypted) 🔒

PERF_CTL[0..5]

[0]	branches
[1]	instructions
[2]	cycles
[1] … [5]	empty

VMCB Control area

ALLOWED_PERF_CTL[0..5]

[0]	branches
[1] … [5]	Empty





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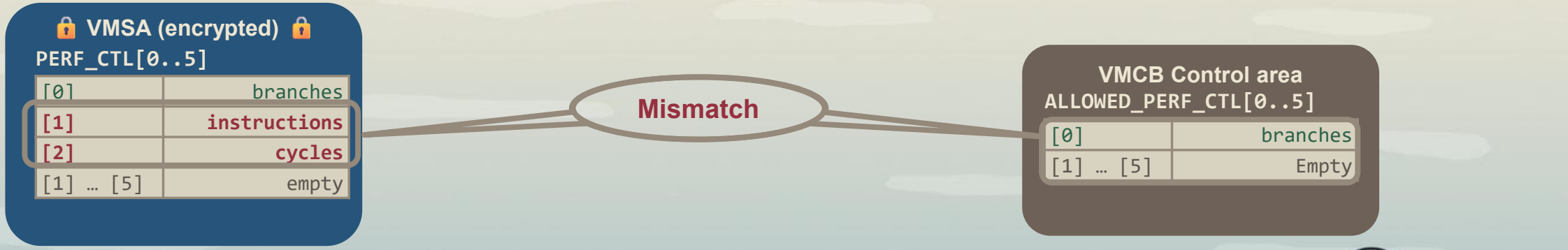
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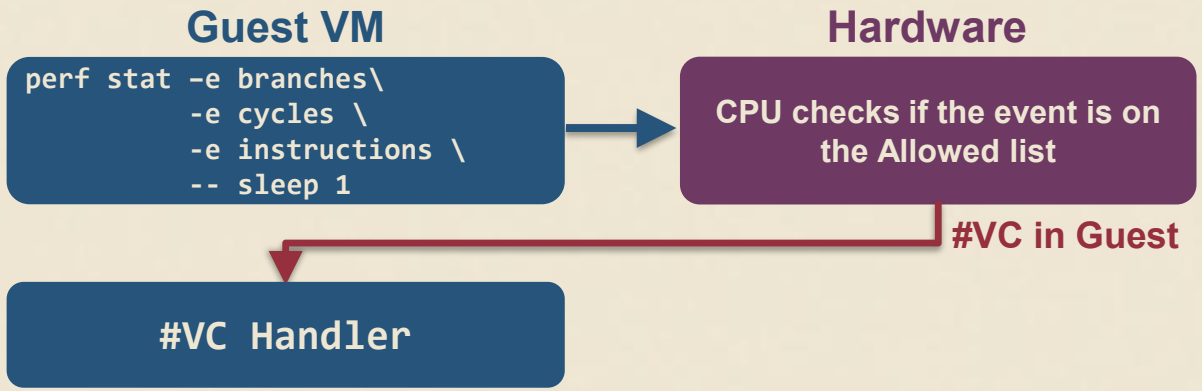
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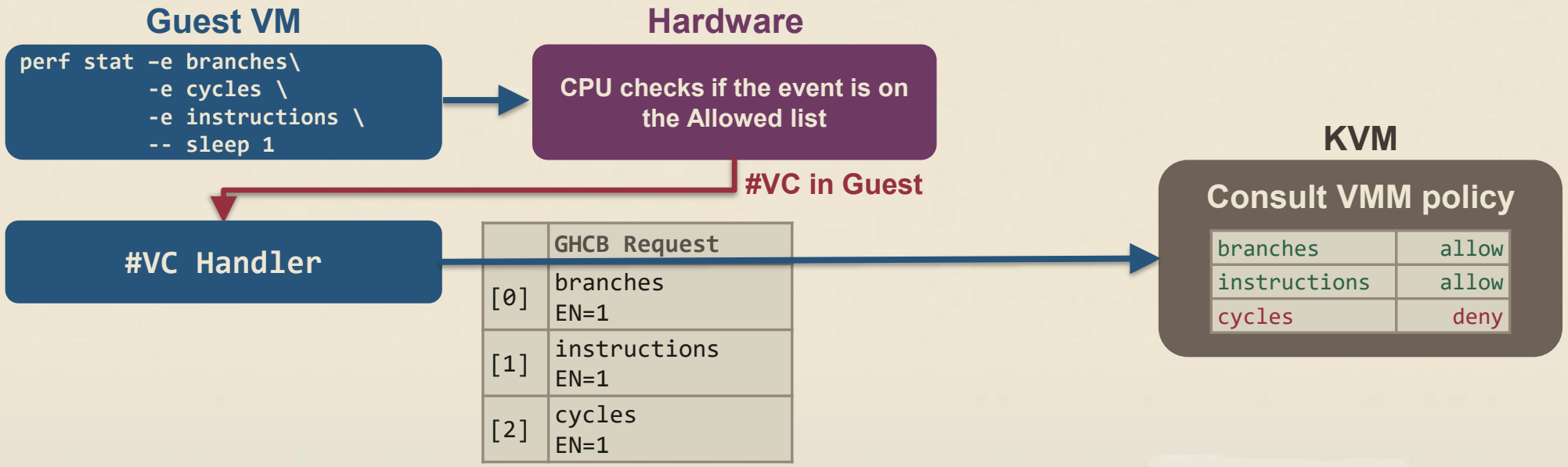


Events not in Allowed list: Exit to KVM



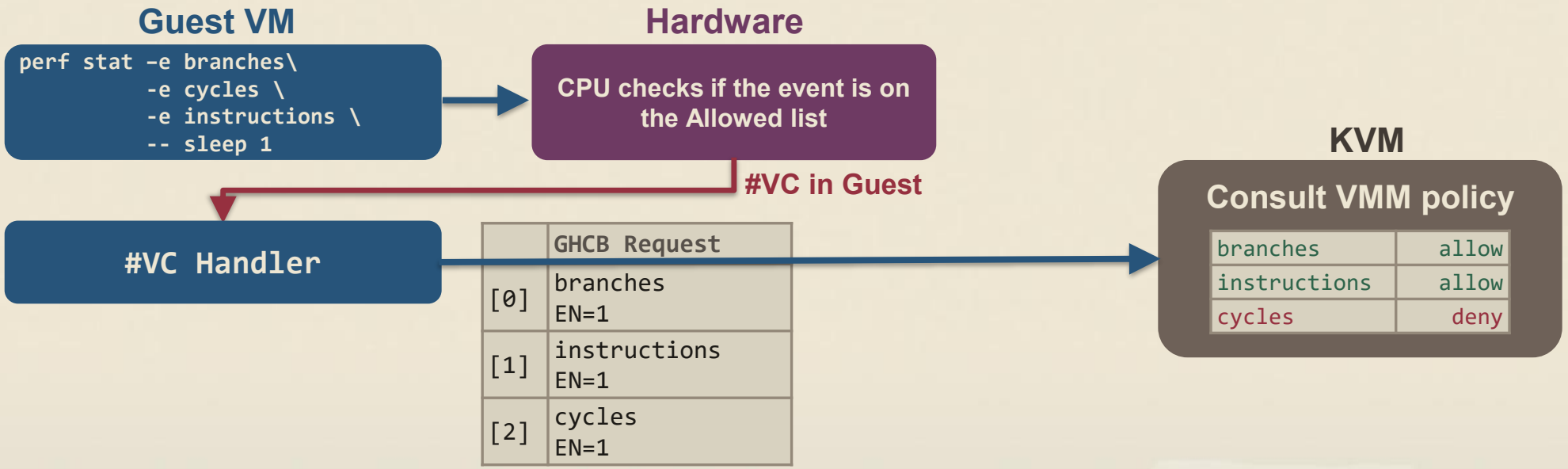


Events not in Allowed list: Exit to KVM





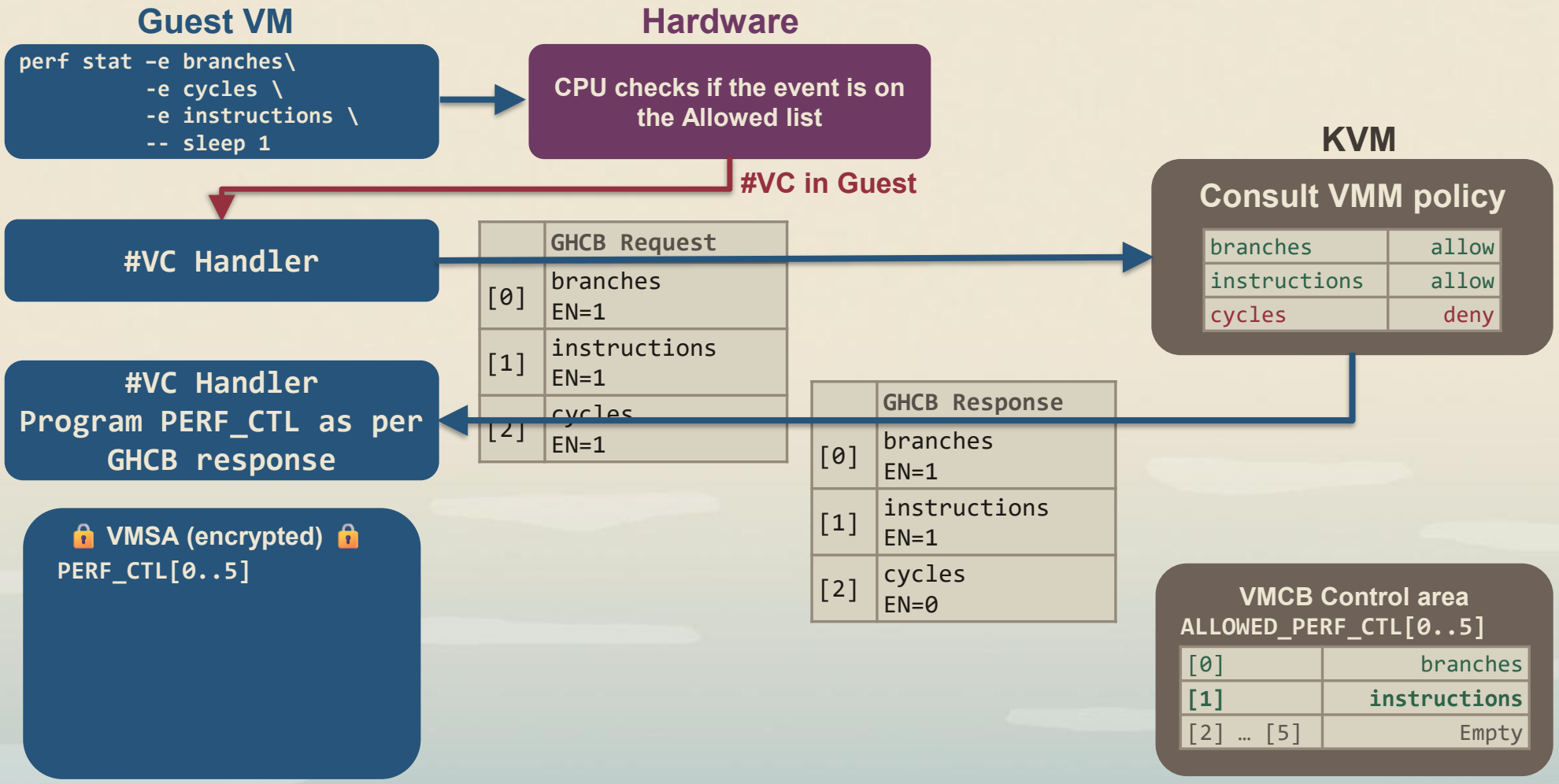
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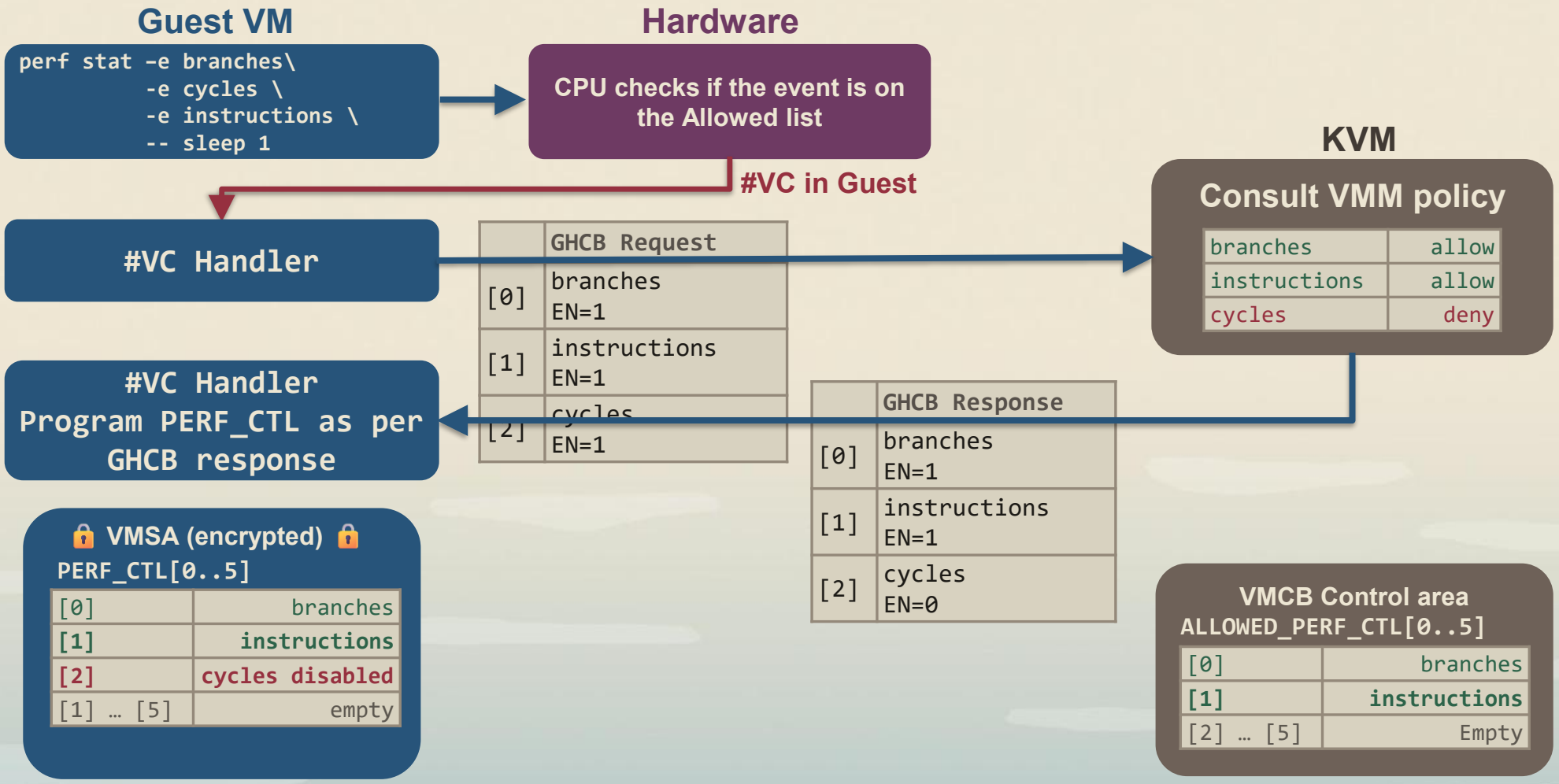
🔒 VMSA (encrypted) 🔒
PERF_CTL[0..5]

VMCB Control area
ALLOWED_PERF_CTL[0..5]

Events not in Allowed list: Exit to KVM



Events not in Allowed list: Exit to KVM



GHCB Envelope

GHCB shared buffer — struct perf_ctls

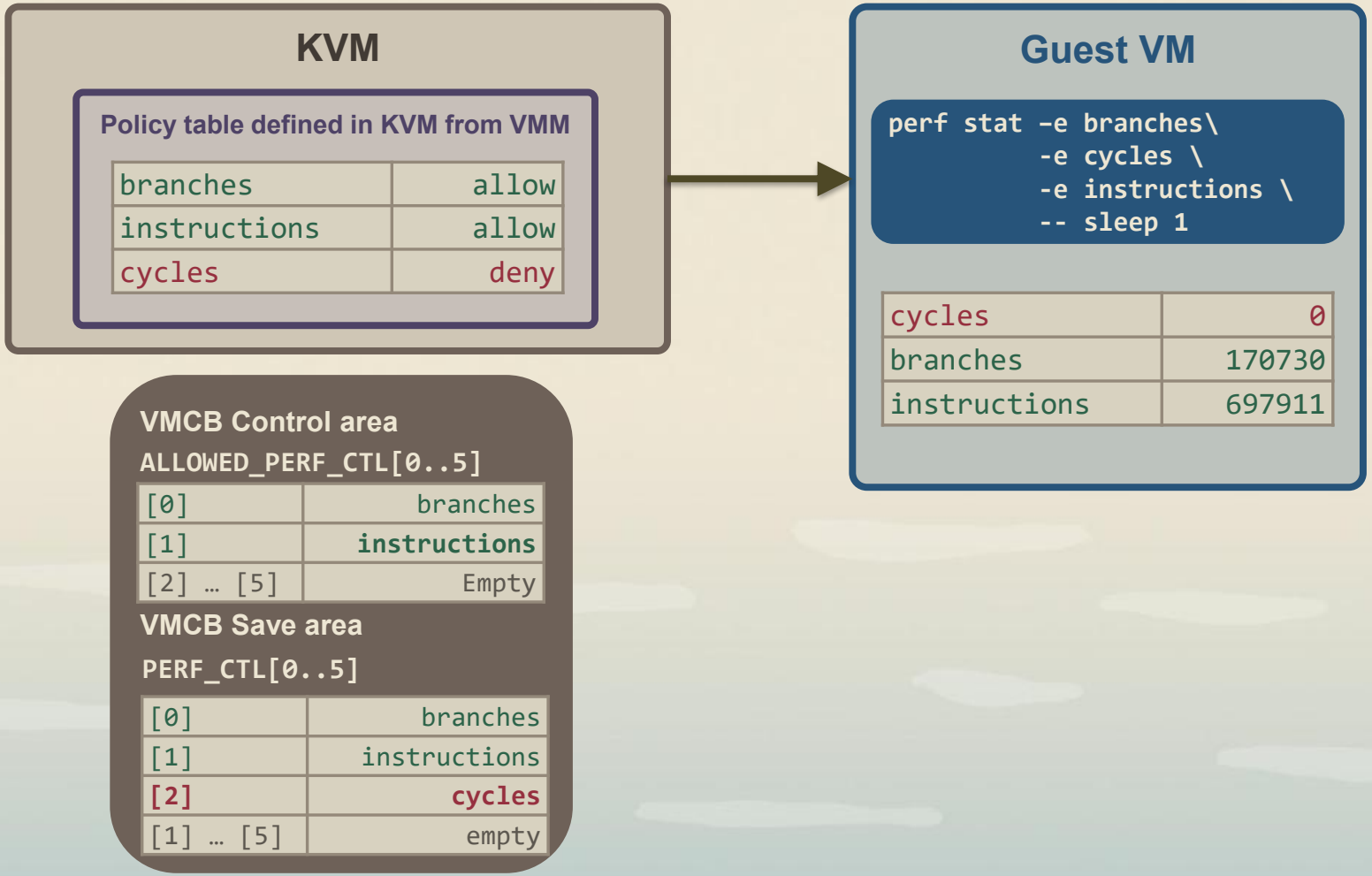
```
struct perf_ctl_entry {
    u64 perf_ctl_idx;   /* MSR index of the event selector */
    u64 perf_ctl_val;   /* its current value */
} __packed;             /* kvm writes the filtered value back  */

struct perf_ctls {
    u8  num_counters;   /* No of counters available            */
    u8  reserved[7];
    struct perf_ctl_entry counters[];
} __packed;
```

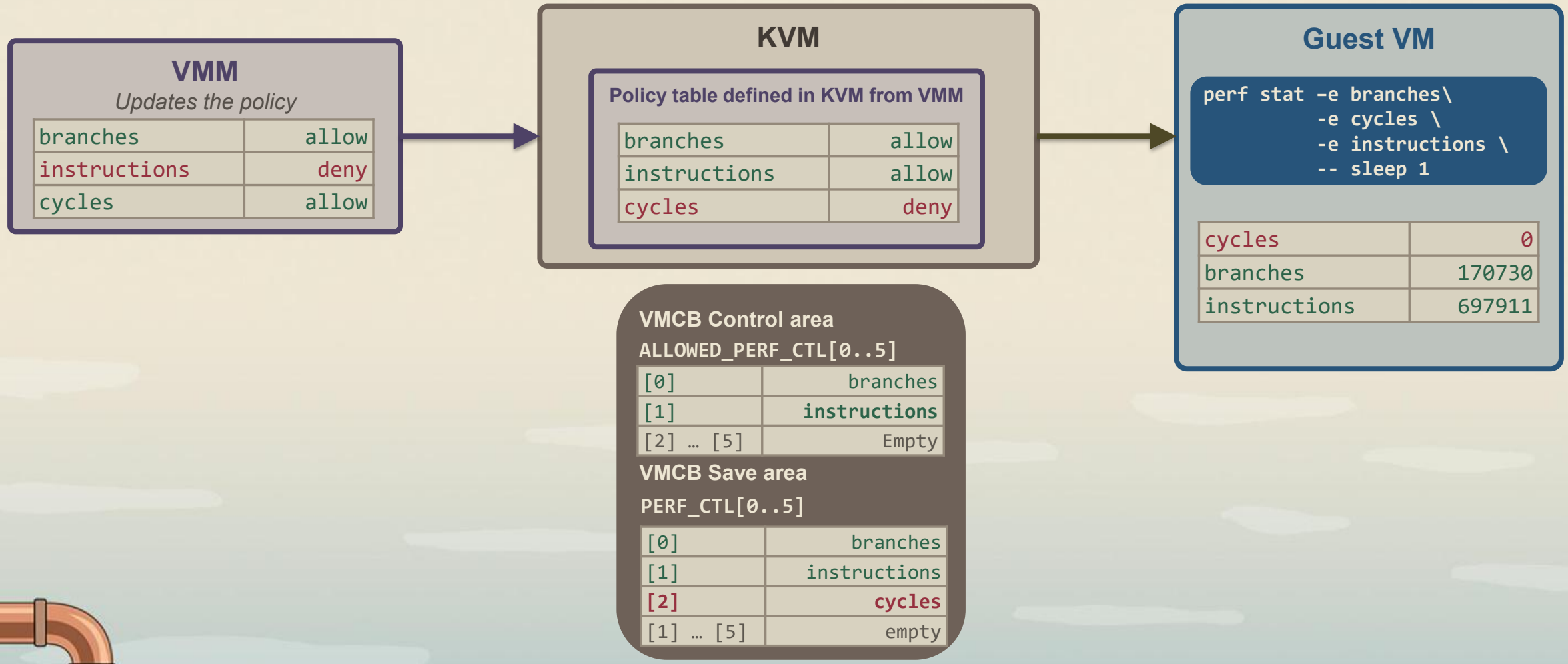
	GHCB Request
SW_EXITCODE	New GHCB Exit code
SW_EXITINFO1/2	0/0
RCX	MSR which is intercepted
EDX:EAX	Value of the MSR
SW_SCRATCH	GPA of GHCB shared buffer

	GHCB Response
SW_EXITCODE	Unchanged
SW_EXITINFO1/2	Success 0/0 / Error code
RCX	Unchanged
EDX:EAX	Filtered Value of the MSR
SW_SCRATCH	Filtered values in SW_SCRATCH area

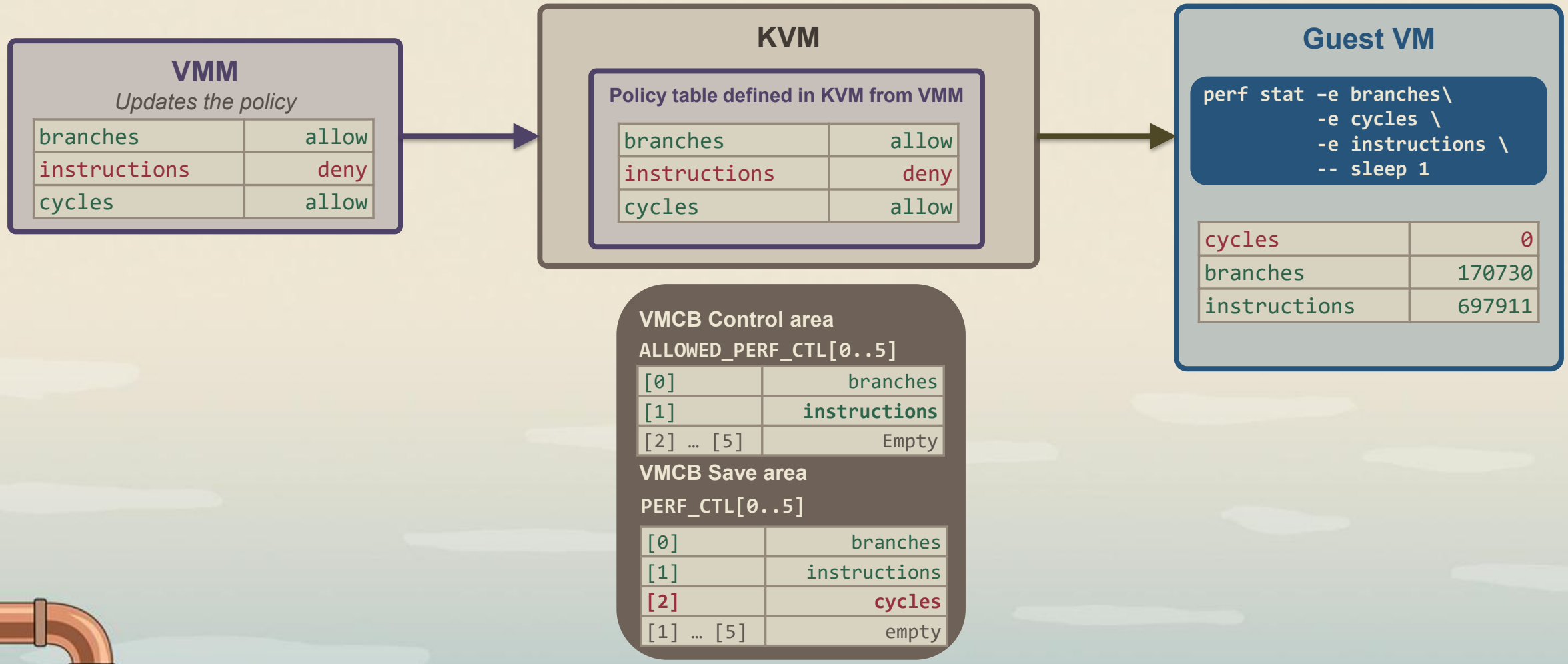
Changing the Filter Mid-Run



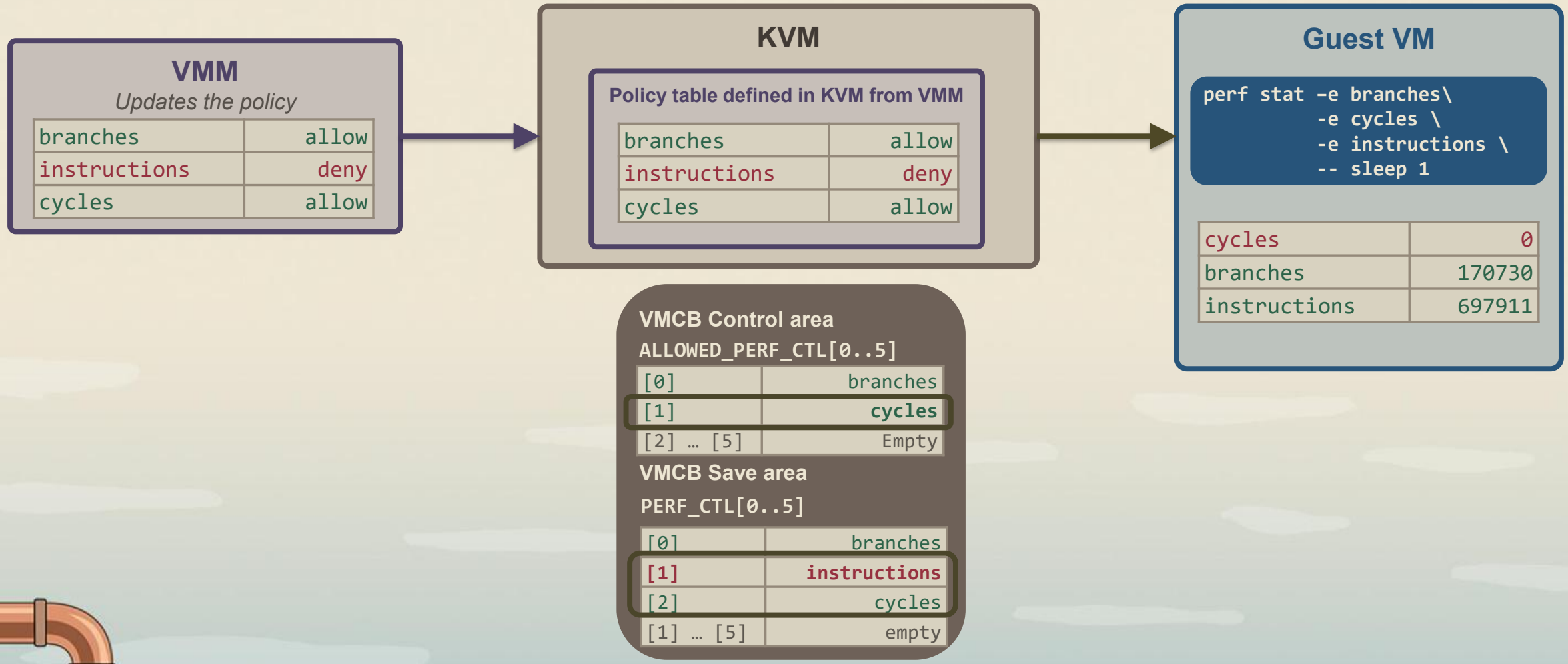
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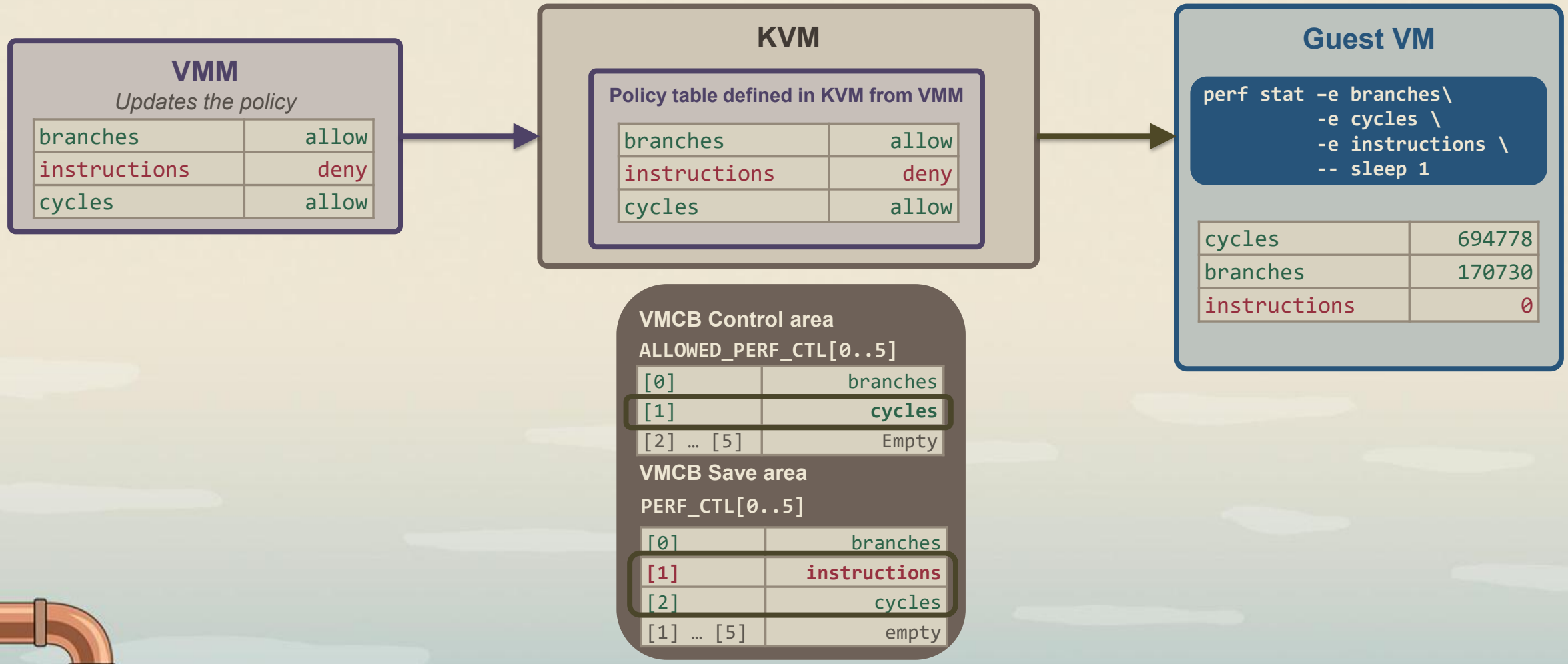


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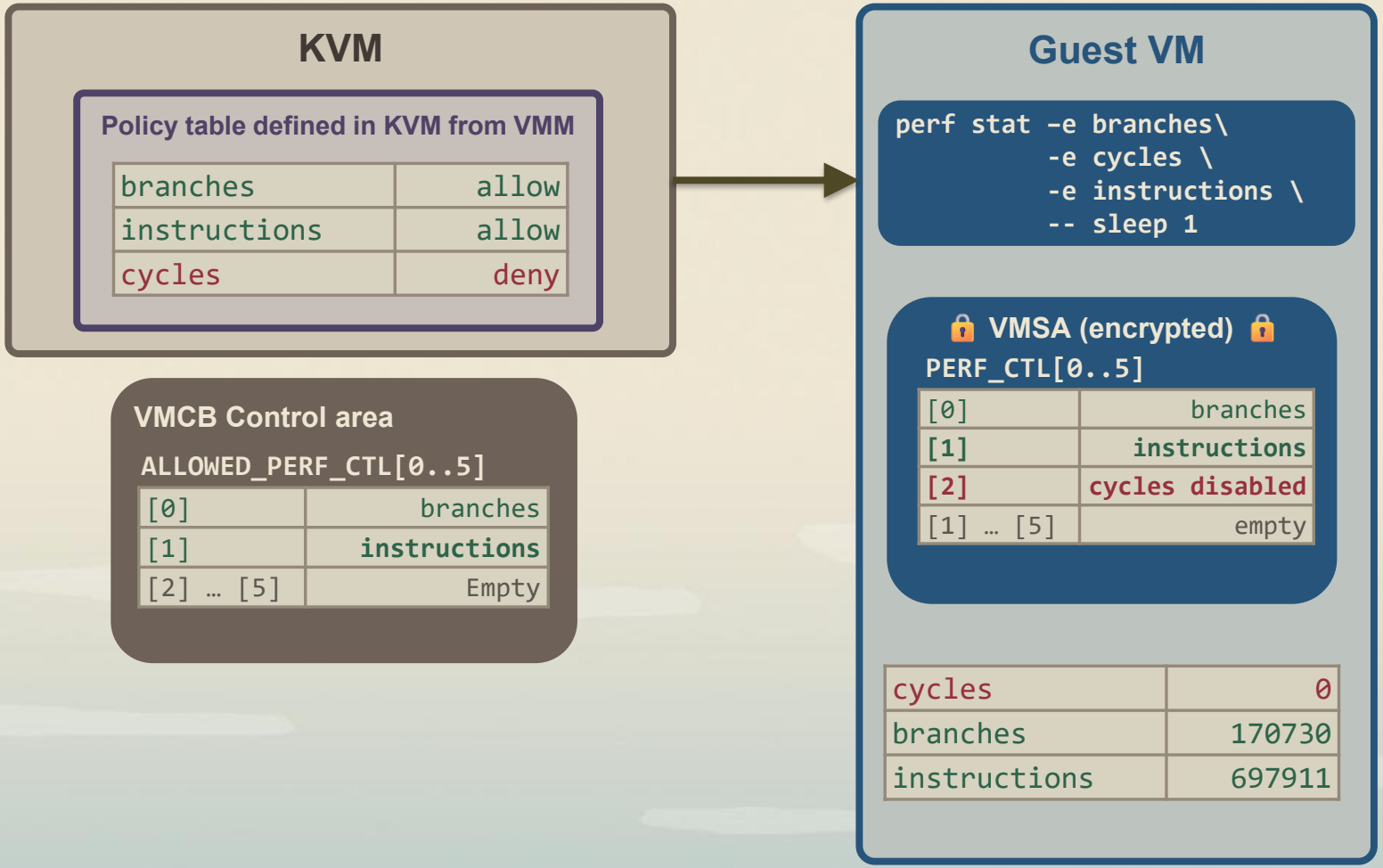


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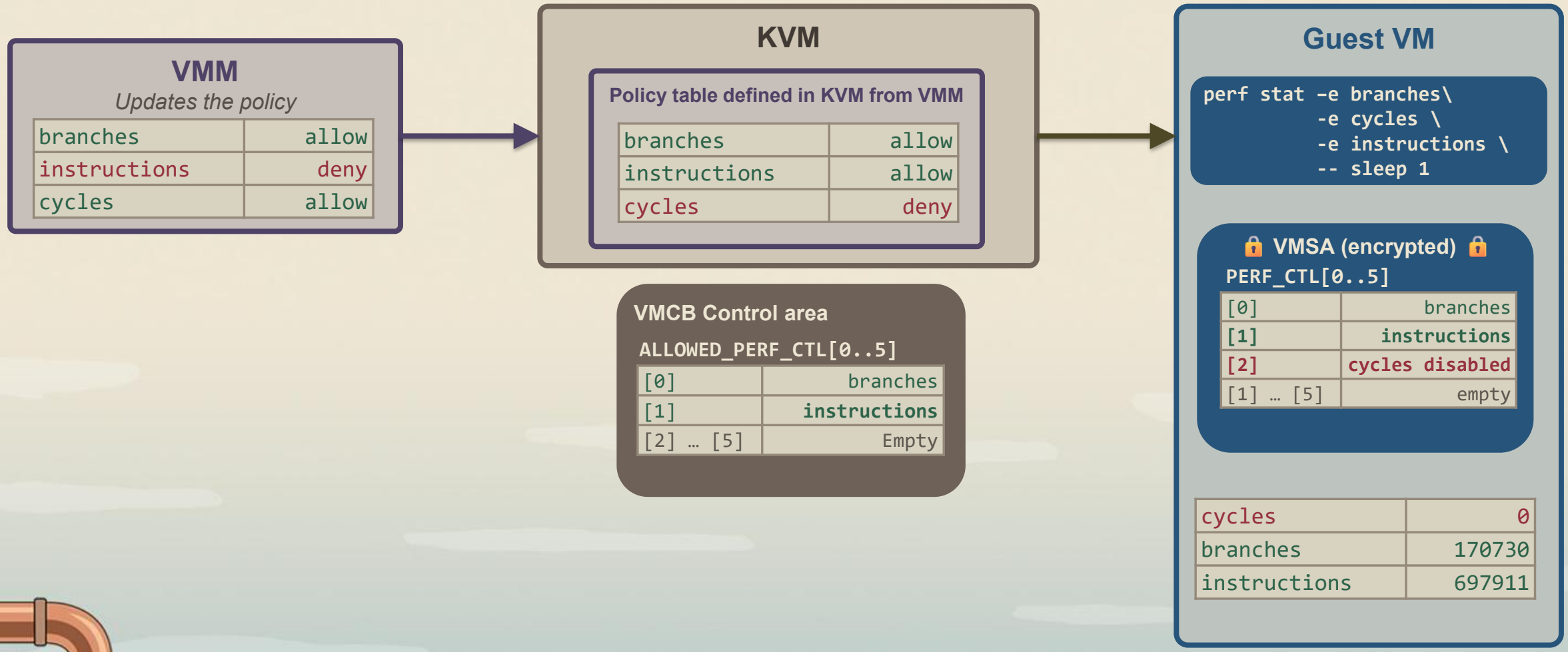
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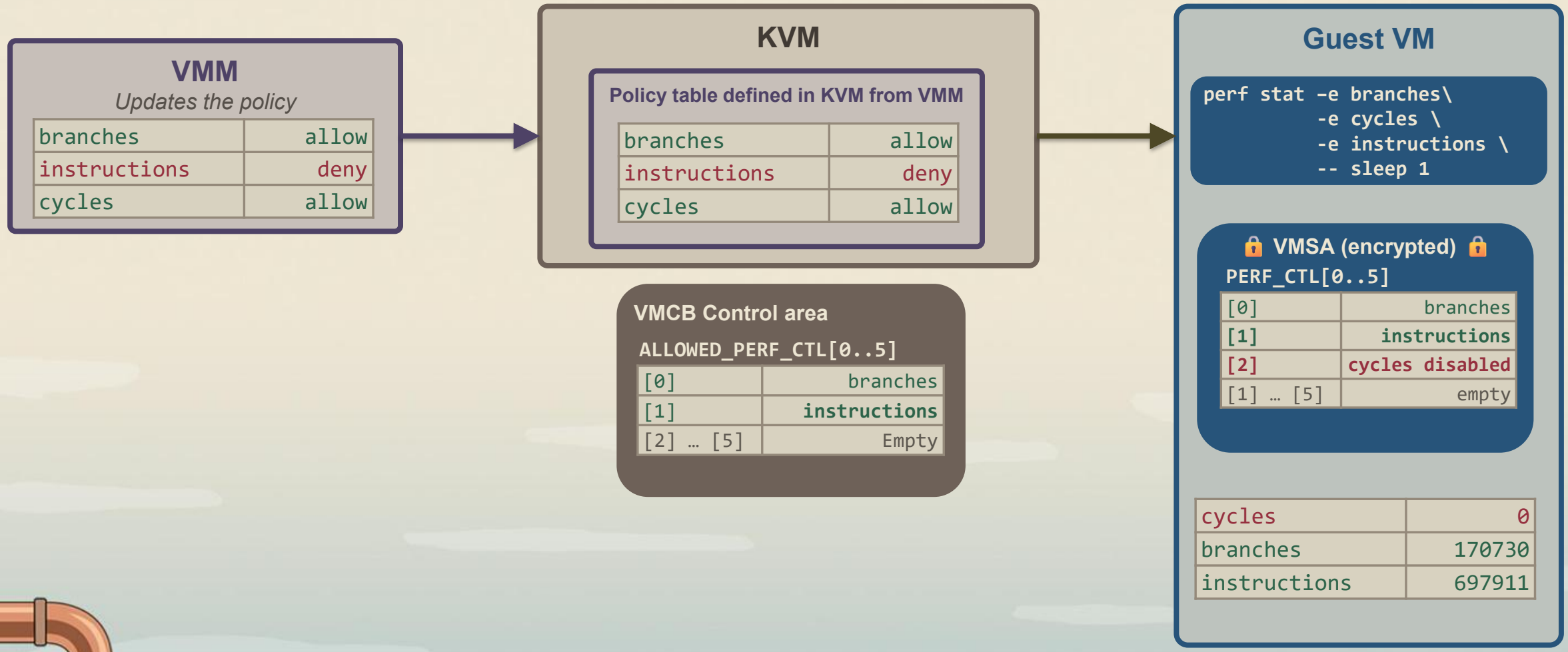
Changing the Filter Mid-Run (Confidential Guests)



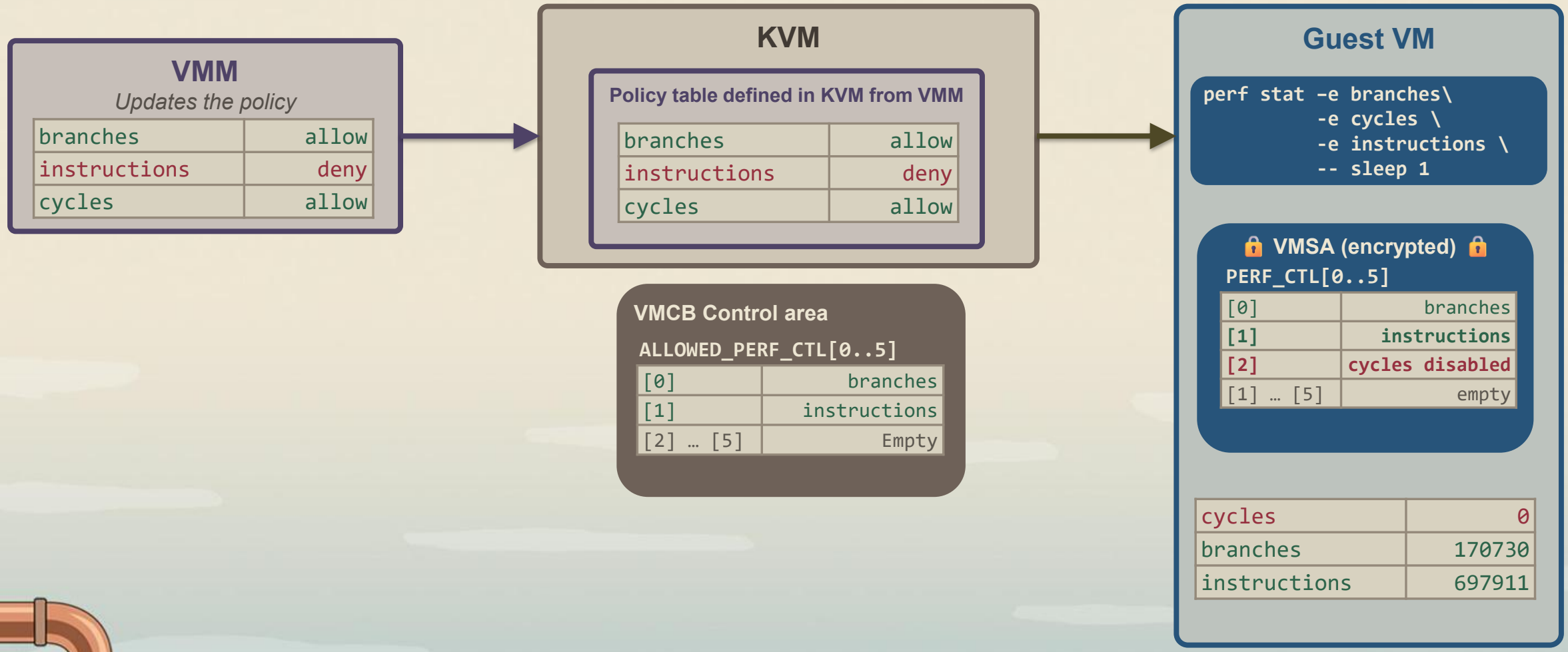
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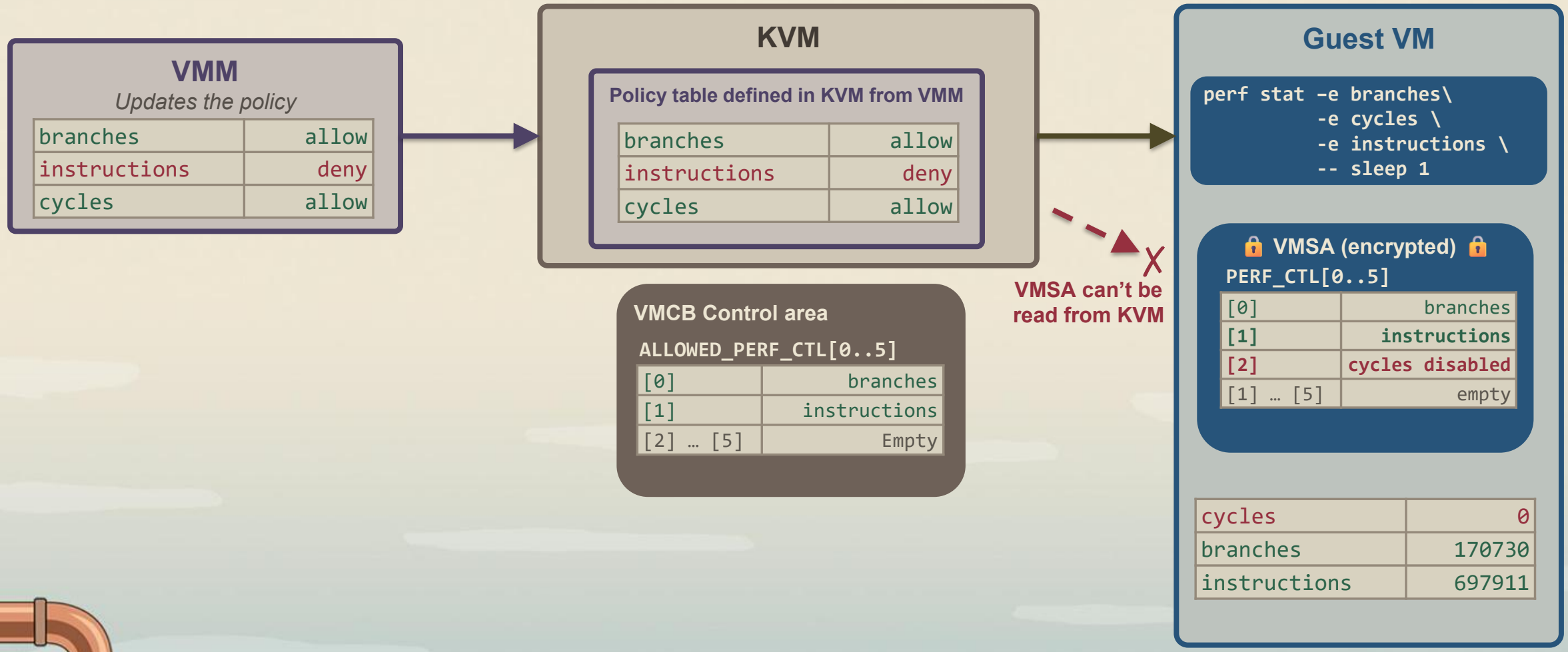
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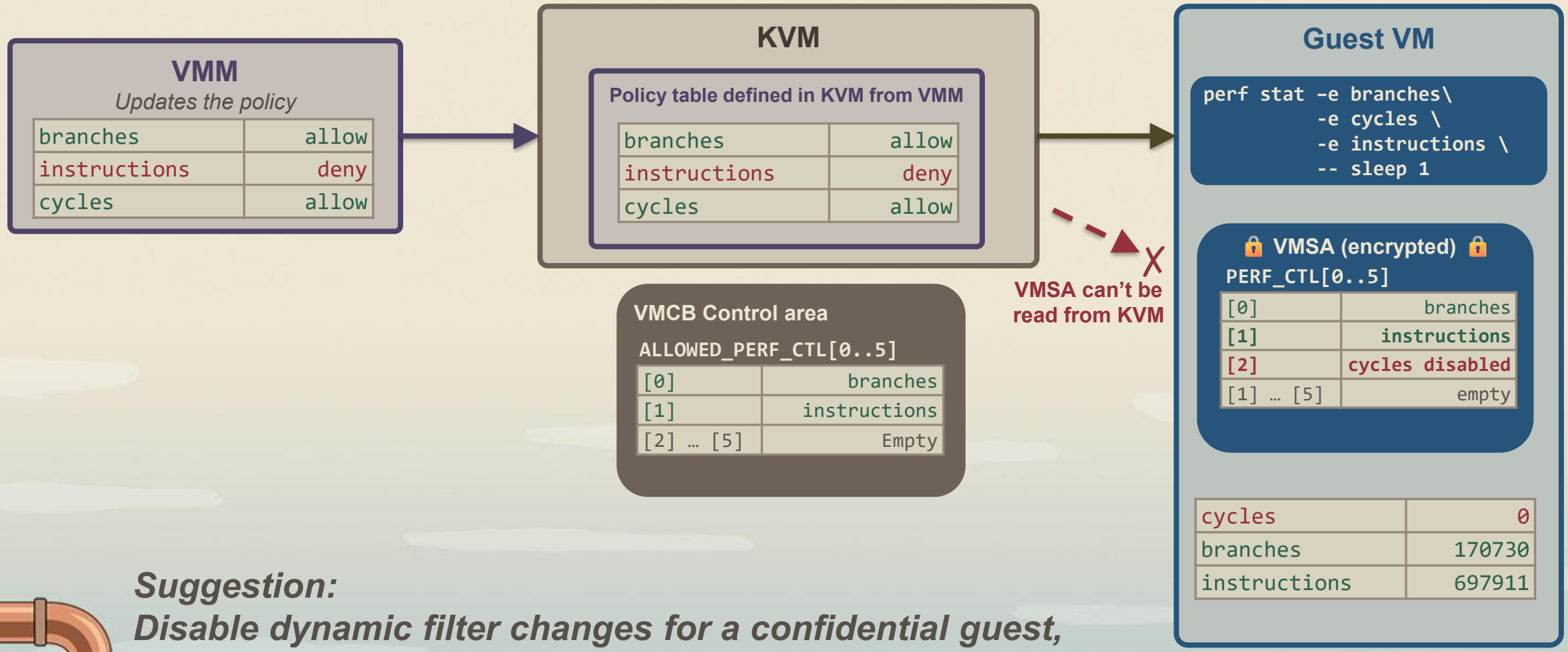
Changing the Filter Mid-Run (Confidential Guests)



Changing the Filter Mid-Run (Confidential Guests)



Changing the Filter Mid-Run (Confidential Guests)



Suggestion:
Disable dynamic filter changes for a confidential guest, fix the filter at guest creation

Current Status

Hardware-Assisted PMC filtering for SVM guests
https://lore.kernel.org/kvm/20261001044351.50535-1-manali.shukla@amd.com/T/#t

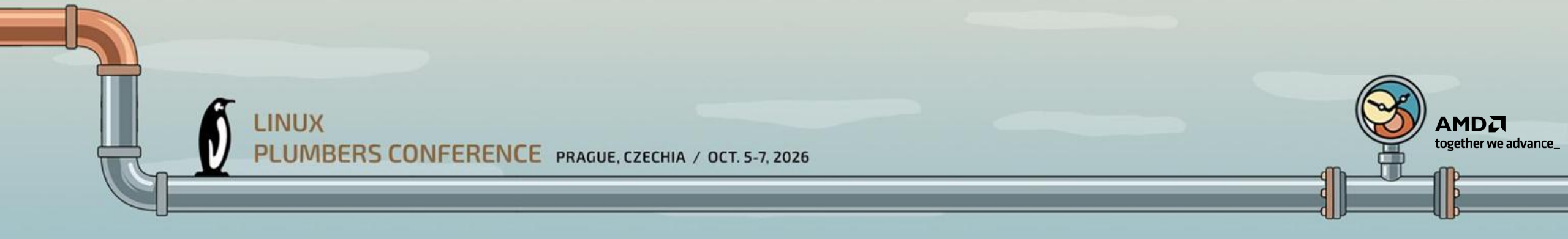
Hardware-Assisted PMC filtering for confidential guests
https://lore.kernel.org/kvm/20261001132021.254270-1-manali.shukla@amd.com/T/#t



References

- AMD64 Architecture Programmer's Manual
  https://docs.amd.com/v/u/en-US/40332_4.09_APM_PUB

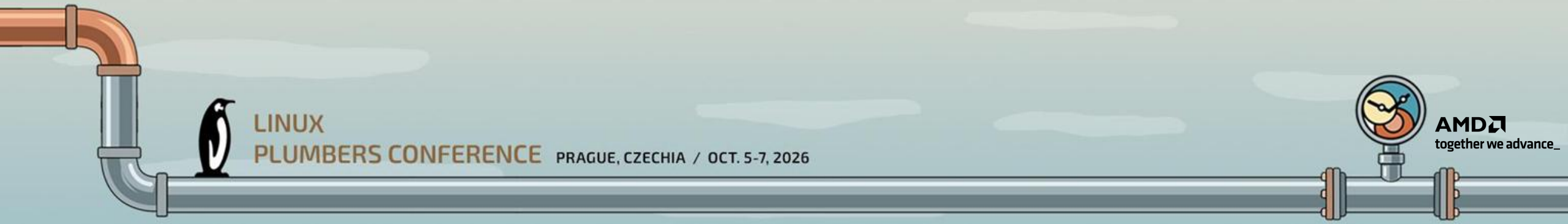
- AMD64 Guest PMC Event Filtering
  https://docs.amd.com/api/khub/documents/2eieIP7KVNk7sKFJ9Ew~Ng/content



Acknowledgements

Nikunj Dadhania

David Kaplan



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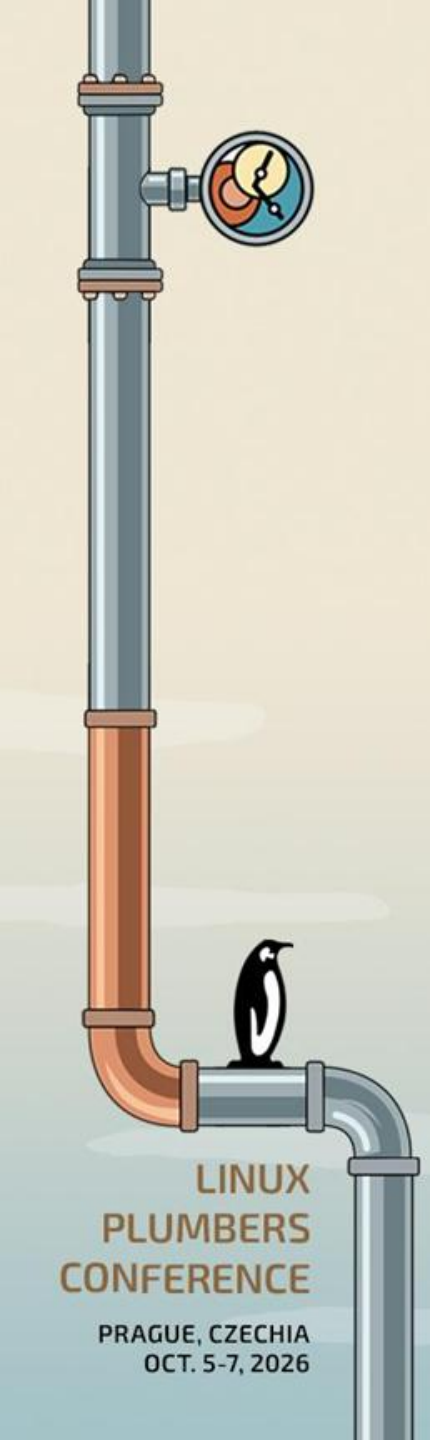
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